Phase-Locked Loop (PLL) Design in Cadence Virtuoso

Progress Report: schematic design and first transient bring-up

45 nm CMOS | Cadence Virtuoso | Spectre transient simulation

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1. Overview

This report covers where we are with the PLL we have been building in Cadence Virtuoso. The aim is a complete charge-pump PLL assembled from its own sub-blocks rather than ideal models: a phase-frequency detector, a charge pump, a passive loop filter, a current-starved VCO, and a frequency divider in the feedback path. Everything is drawn and simulated at the transistor and gate level in a 45 nm process.

At this point every block has a finished schematic, the blocks are wired together in a single top-level cell, and the full loop runs through a Spectre transient simulation. The sections below walk through each block, show what the first full-loop simulation looks like, and set out what we plan to tackle next.

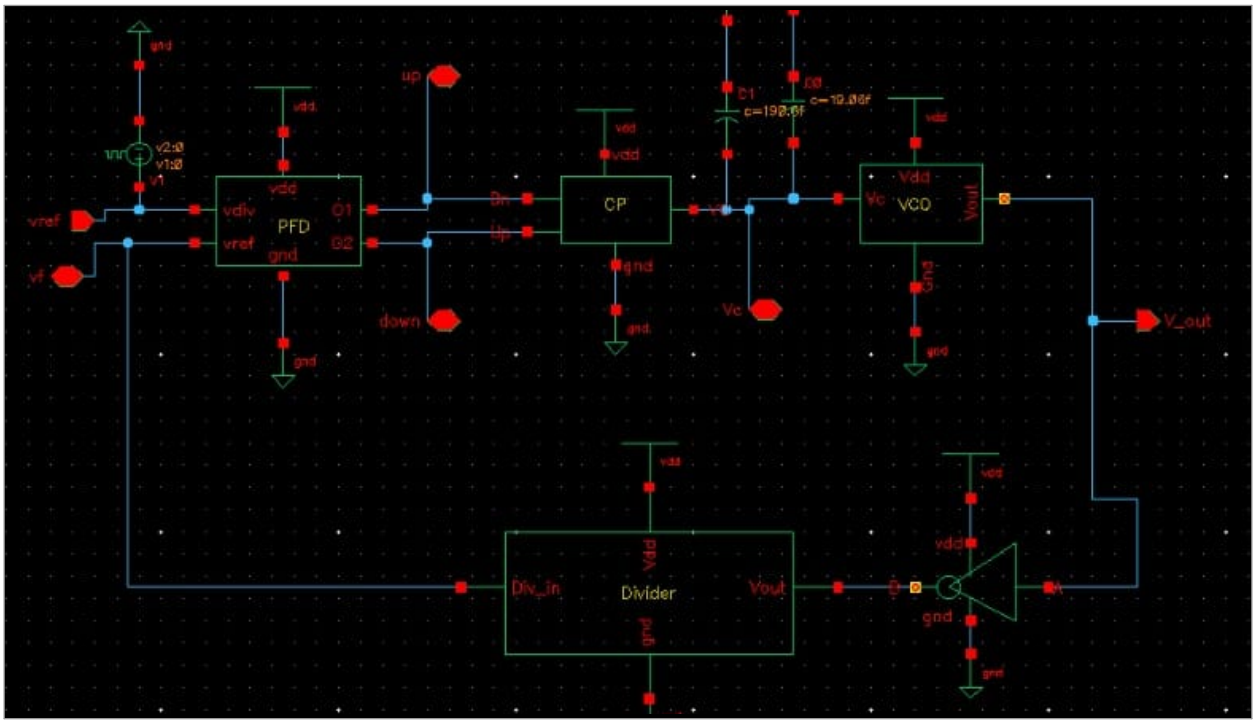


Figure 1. PLL top-level schematic: PFD, charge pump, loop filter, VCO, divider and feedback inverter.

2. Design Progress

2.1 Top-level architecture

The top-level cell ties the five functional blocks together with a clean set of probe points: the reference and feedback inputs, the up and down pulses from the detector, the control voltage Vc, and the VCO output V_out. Keeping those nets exposed from the start made it easy to debug block interaction later, and they are the same signals we plot in the simulation section.

2.2 Phase-frequency detector

The PFD is built from two resettable D flip-flops and a two-input gate that feeds their reset pins, which gives the usual up/down pulse behaviour and removes the dead zone. Both flip-flops take their clocks from the reference and the divided feedback signal, and the two outputs go straight to the charge pump.

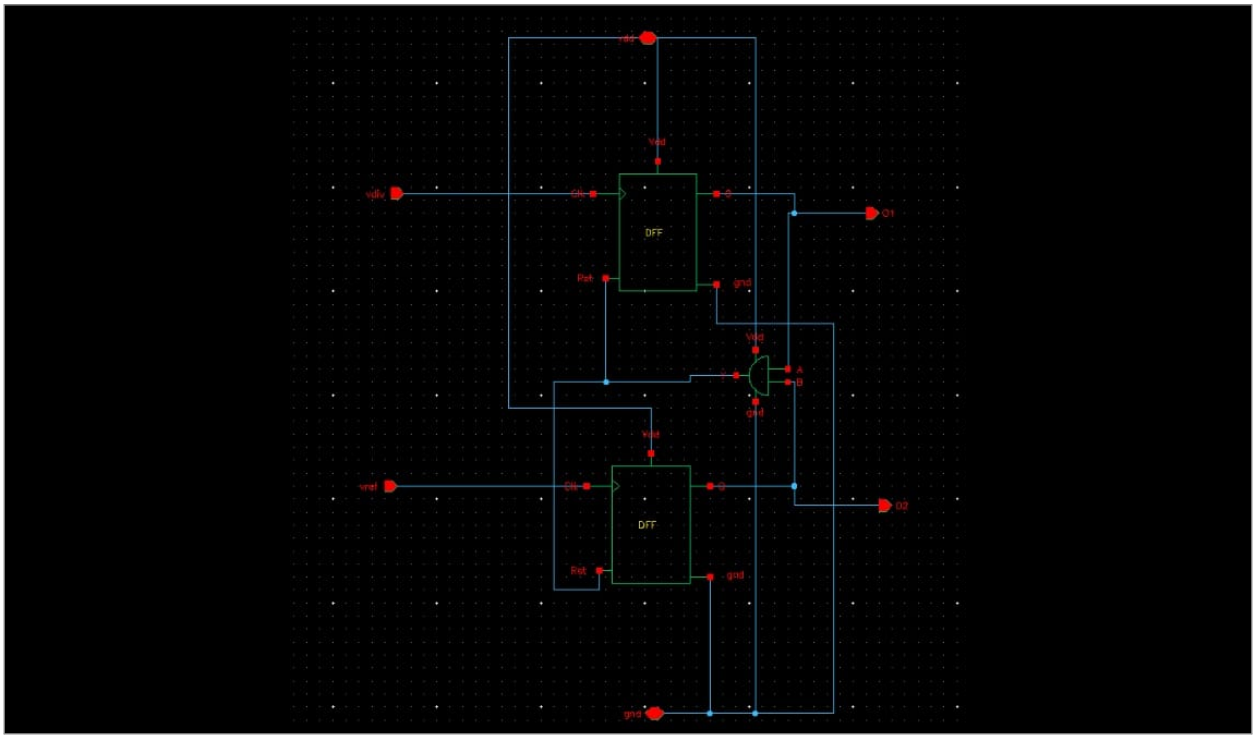


Figure 2. PFD schematic with two resettable flip-flops and the reset gate.

The flip-flop with reset is a gate-level design made from four NOR gates. Building it this way, instead of dropping in a library cell, gave us direct control over the reset path and kept the whole detector self-contained.

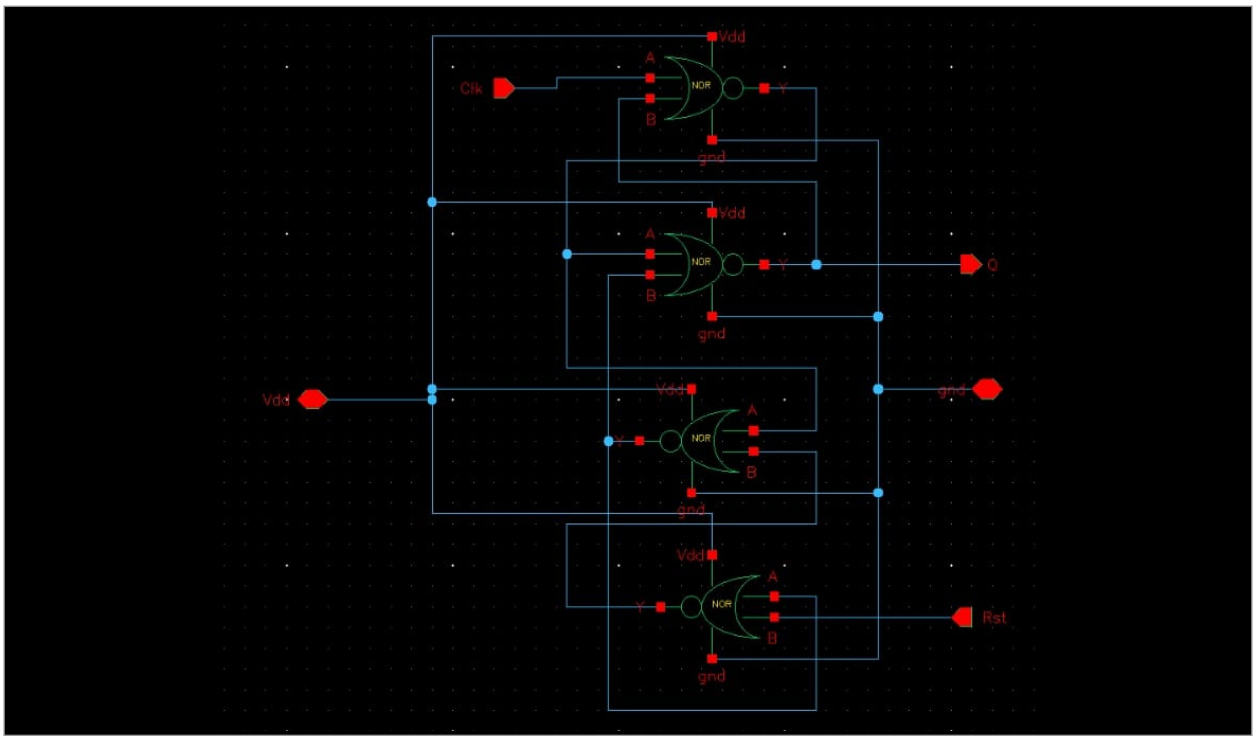


Figure 3. Resettable D flip-flop (DFF_rst) built from NOR gates.

2.3 Charge pump

The charge pump uses a PMOS pull-up branch and an NMOS pull-down branch, both switched by the detector outputs, with a 15 µA bias current setting the pump current through current mirrors. An inverter on the Up path takes care of the polarity of the PMOS switch. All devices are 45 nm transistors drawn at W = 120 nm and L = 45 nm.

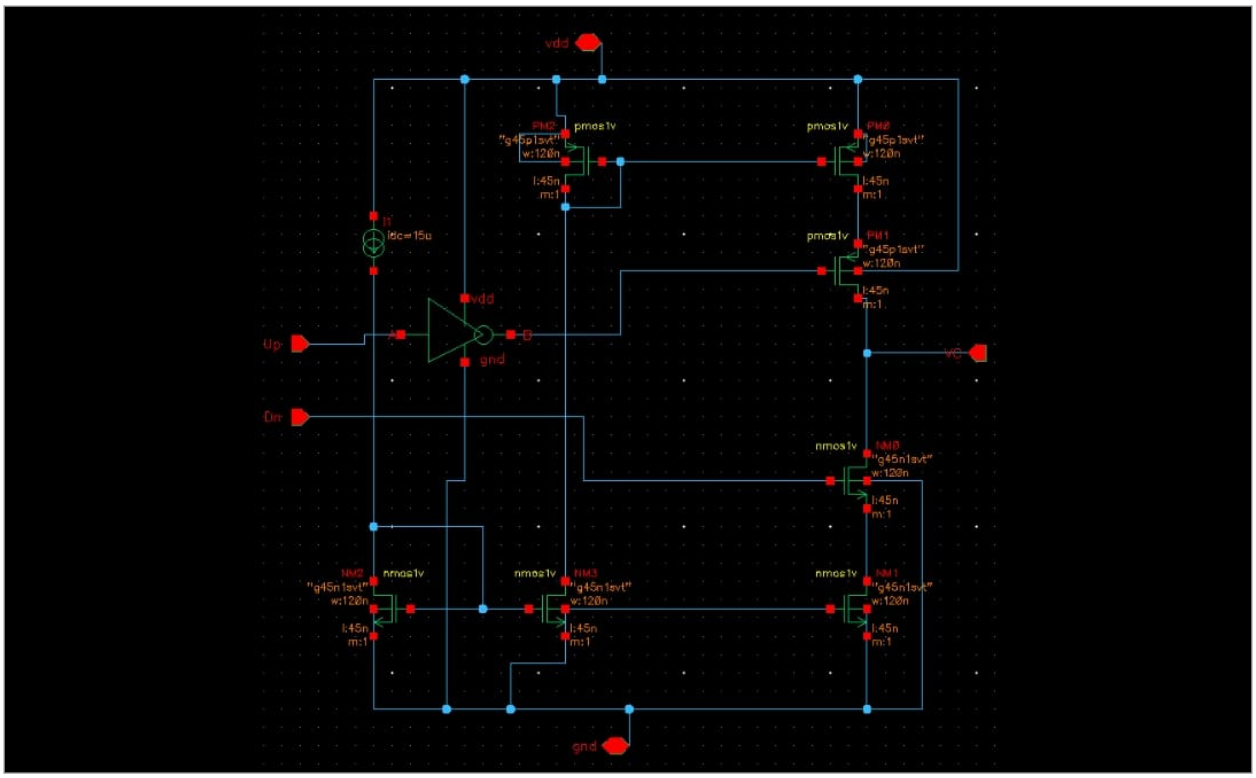


Figure 4. Charge pump schematic with mirror branches and Up/Down switching.

2.4 Loop filter

The loop filter is a passive second-order network: a series resistor and capacitor (R1 and C1) with a smaller shunt capacitor (C2) on the control node. The resistor adds the stabilising zero, and the small capacitor smooths the ripple on Vc before it reaches the VCO. [Confirm and insert final values: R1 = ___ kohm, C1 = ___ fF, C2 = ___ fF.] It sits in the top-level schematic, visible in Figure 1.

2.5 Voltage-controlled oscillator

The VCO is a current-starved design. The control voltage sets the bias currents through a mirror network, and these in turn set the delay of the oscillator stages, so Vc maps onto output frequency. The schematic comes to 29 instances, including the bias resistors and an output stage that drives V_out.

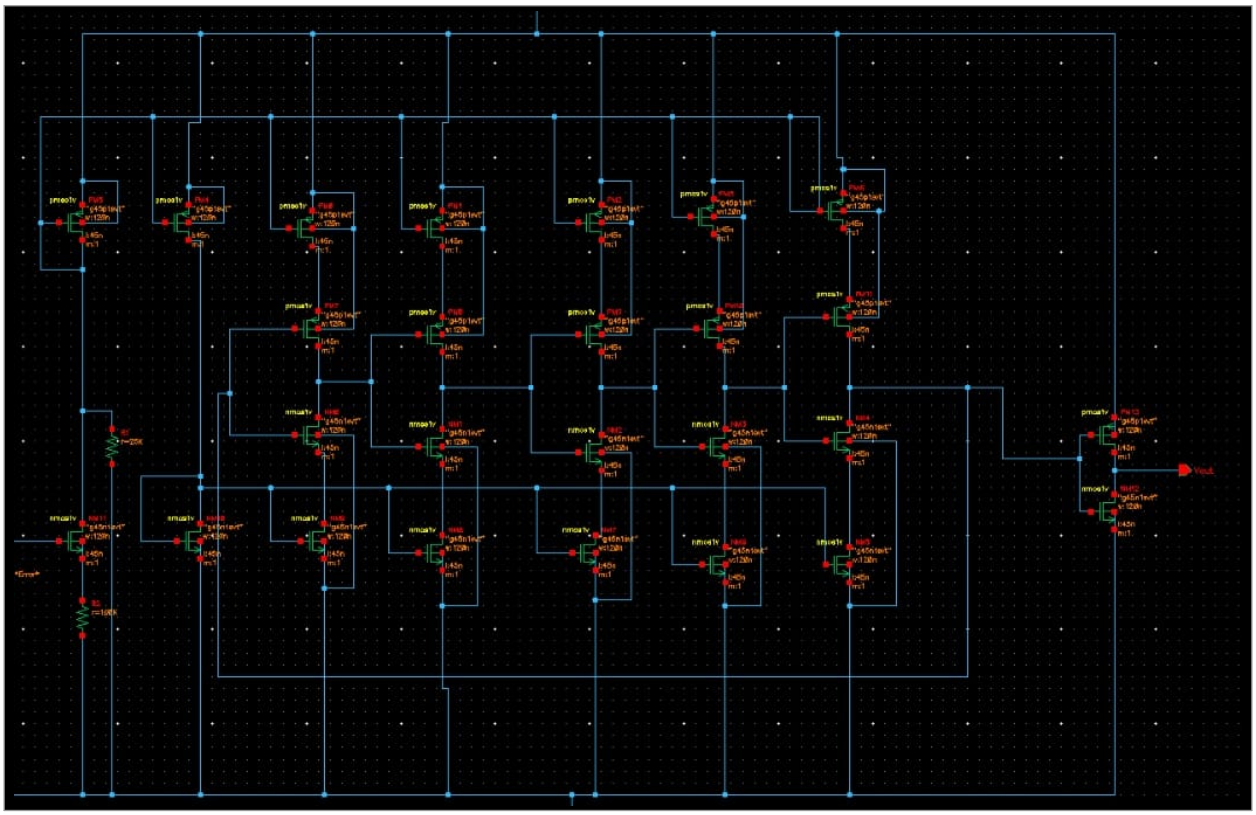


Figure 5. Current-starved VCO schematic.

2.6 Frequency divider

The divider is a chain of six TSPC (true single-phase clock) flip-flops with a gate that shapes the division ratio. TSPC was a deliberate choice: it needs only a single clock phase, keeps the transistor count low, and works well at the speeds we are targeting.

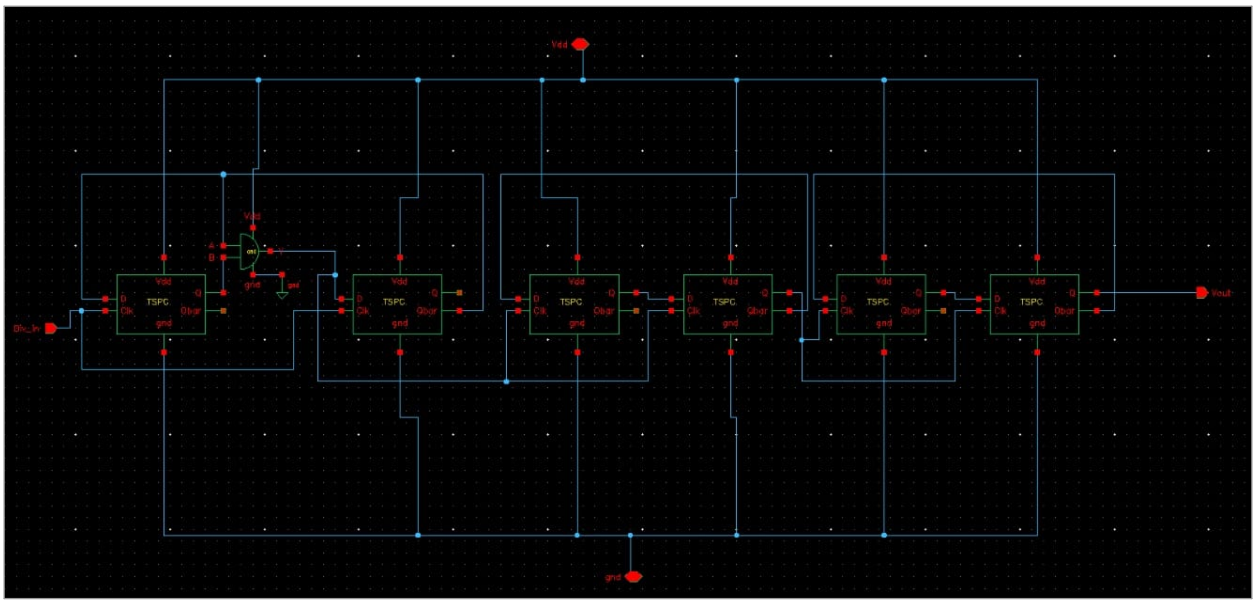


Figure 6. Frequency divider built from cascaded TSPC flip-flops.

2.7 Feedback inverter

A simple CMOS inverter, again with 45 nm devices at W = 120 nm and L = 45 nm, sits in the feedback path and sharpens the edges that reach the divider and detector.

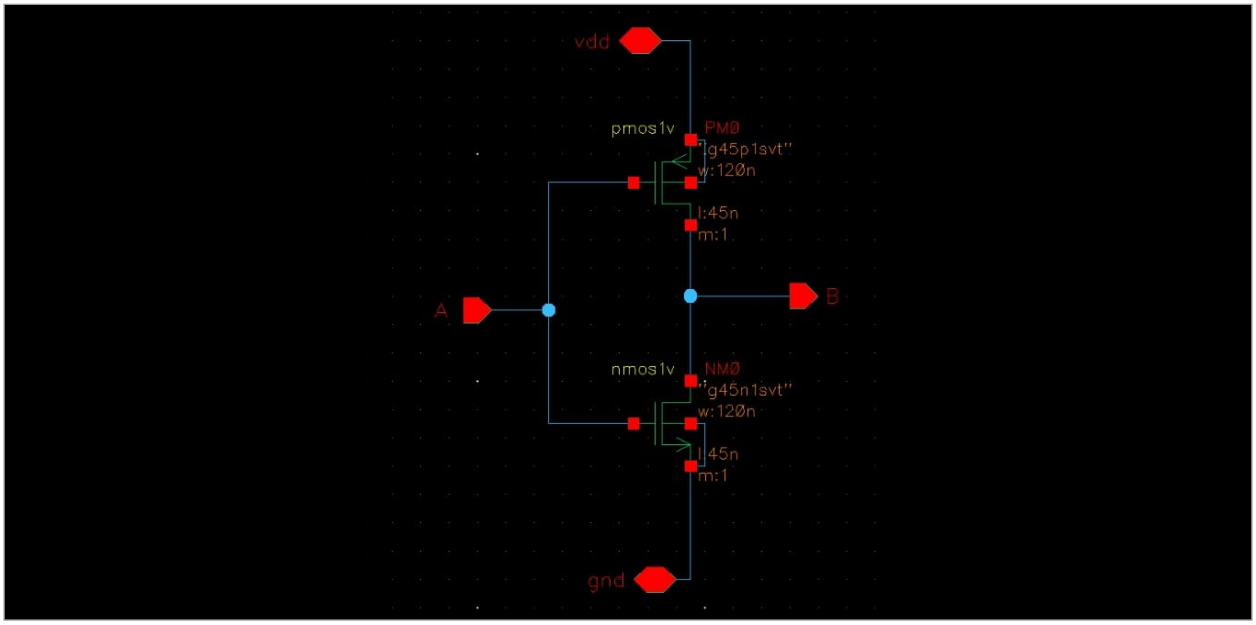


Figure 7. CMOS inverter used in the feedback path.

3. Simulation Results

With the blocks integrated, we ran the first full-loop transient in Spectre over 1.5 µs. We probed the reference, the feedback signal, the up and down pulses, the control voltage, the VCO output, and the branch currents through one PMOS and one NMOS device in the charge pump. Figure 8 shows the results.

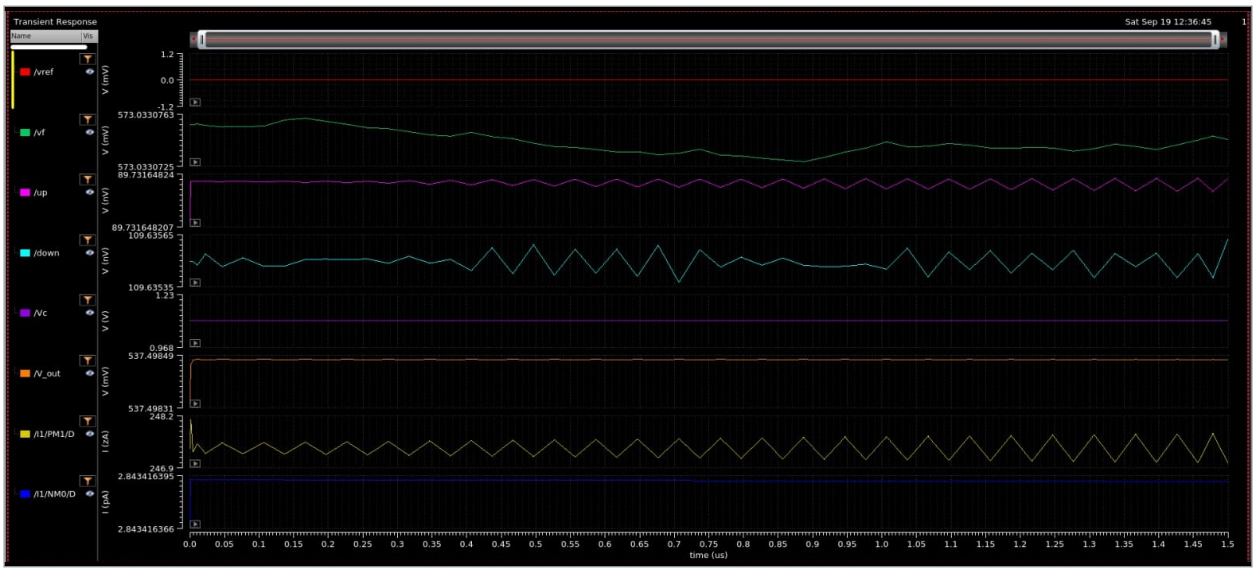


Figure 8. Transient response of the PLL (0 to 1.5 µs): vref, vf, up, down, Vc, V_out and charge-pump branch currents.

The run completed cleanly and the loop settles within the first few nanoseconds to a steady operating point. From there V_out holds a stable level for the rest of the window and the control voltage stays flat, with no drift over the full 1.5 µs. The up and down outputs and the charge-pump branch currents show a regular, repeating ripple, which tells us the detector and pump are active and interacting the way we expect.

One thing worth keeping in mind when reading the plot: the viewer autoscales each trace independently, so the small up/down excursions and the pump-current ripple look much larger on screen than they are. Read the axis values and they are fine structure sitting on top of a very steady bias point.

4. Key Observations

The main outcome is that the whole hierarchy works together. Six separately designed cells netlist into one top level, the simulator converges without trouble, and the loop reaches a stable, repeatable starting state. That is a solid base for everything that comes next.

Vc is steady across the run, which is what we want from the loop filter and charge pump at this stage. The up and down signals stay balanced in scale, and the pump currents track the detector activity, so the PFD-to-charge-pump handoff is behaving as designed.

This first run was a bring-up check on the bias and signal path. The reference was held at a fixed level, so it exercised the biasing and block interaction rather than a full lock sequence with a toggling reference. Setting up the clocked reference is the natural next move, and the probe points we have already placed will carry straight over.

5. Summary of Completed Work

Here is where each piece stands today.

Block	Status
PFD and resettable DFF	Schematic complete; DFF built from NOR gates
Charge pump	Schematic complete; 15 µA mirror bias, 45 nm devices
Loop filter	Second-order R-C network in place at top level
VCO (current-starved)	Schematic complete and integrated
Divider (TSPC chain)	Schematic complete and integrated
Feedback inverter	Schematic complete and integrated
Top-level PLL	Integrated; 1.5 µs transient runs cleanly and settles to a stable operating point

6. Next Steps

The next stage is to move from bring-up to proper lock behaviour. We will drive the reference with a clocked source at the target frequency and run a longer transient to watch lock acquisition and measure lock time. In parallel we plan a Vc sweep on the VCO to produce its tuning curve and confirm the gain and range, then use those numbers to fine-tune the loop filter for damping and bandwidth.

After that, we will look at jitter and output spectrum, check behaviour across process, voltage and temperature corners, and start preparing for layout.